if an element is in the heap (or get its position) in constant time, on average.

Listing 2.10 The contains method

```
function contains(elem)
  index ← elementToIndex[elem]
  return index >= 0
```

The function uses an extra field `elementToIndex` we can add to our heap: this is possible under 2 assumptions:

1. That `elementToIndex[elem]` by default returns `-1` if `elem` is not stored in the heap;
2. That we don't allow duplicate keys in the heap (otherwise, we will need to store a list of indices for each key).

2.6.9 Performance recap

With the description of the `contains` method, we have concluded our overview of this data structure.

We have presented several operations on heaps, so that it feels the right time to make order and recap their running time, and also – just as importantly – the amount of extra memory they need.

Table 2.4 Operations provided by heaps, and their cost on a heap with n